CREDIT CARD FRAUD DETECTION

PROJECT REPORT

Transaction Analysis • Fraud Monitoring • Power BI Dashboard

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Project focus

Exploratory analysis of genuine and fraudulent credit card transactions, with interactive risk insights.

A concise project document describing the analytical goals, dashboard design, reported metrics, findings, and practical limitations.

Executive Summary

This project examines credit card transaction activity and presents fraud-monitoring insights through a Python analysis notebook and an interactive Microsoft Power BI dashboard. The dashboard is designed to help viewers compare genuine and fraudulent transactions, review fraud rates across transaction amount categories, inspect hourly fraud activity, and understand the distribution of transactions across alert levels.

The dashboard reports **283,726 transactions**, including **473 fraud transactions** and **283,253 genuine transactions**. The overall fraud rate is approximately **0.1667%**, and the reported total fraud amount is **58,591.39**.

Key metric	Reported value
Total transactions	283,726
Fraud transactions	473
Genuine transactions	283,253
Overall fraud rate	0.1667%
Total fraud amount	58,591.39

Important interpretation: this report describes the metrics and visual analysis available in the project. It does not claim that a predictive machine-learning model was trained or evaluated, because model-training details and performance results were not provided.

1. Introduction

Credit card fraud is an important issue for financial institutions, merchants, and customers. Fraudulent transactions can create financial losses, customer inconvenience, and operational costs. Transaction data can be explored to identify patterns and communicate risk-related information in a clear and accessible way.

The purpose of this project is to analyze a credit card transaction dataset and build a dashboard that summarizes transaction volume, fraud activity, and alert-level patterns. The dashboard provides an overview rather than making a real-time authorization decision.

2. Problem Statement

A large volume of credit card transactions can make it difficult to quickly understand how frequently fraud occurs and how fraud activity varies across transaction characteristics. The project addresses this reporting challenge by organizing transaction data into meaningful measures and interactive visualizations.

3. Project Objectives

- Quantify genuine and fraudulent transactions.
- Measure the overall fraud rate and reported fraud amount.
- Compare fraud rates across transaction amount categories.
- Explore the distribution of fraudulent transactions by hour.
- Review transaction counts and fraud rates across alert levels.
- Present the findings in an interactive, easy-to-read Power BI dashboard.

4. Dataset Overview

The project uses a credit card transaction dataset represented in the Power BI model as **credit_card_fraud_analysis**. The fields referenced in the dashboard include:

Field	Role in the analysis
Class	Transaction label used to distinguish genuine (0) and fraud (1).
Amount_Category	Groups transaction amounts into Low, Medium, High, and Very high categories.
Time	Transaction time field available in the data.
Hour	Hour-based field used for hourly analysis and dashboard filtering.

The repository includes the Jupyter Notebook and a dashboard preview image. The original dataset and Power BI (.pbix) file are not included because of their large file sizes. Consequently, this report does not specify the dataset publisher, exact date range, row-level schema, or reproducibility steps that depend on those files.

5. Analytical Approach

The project follows a descriptive-analysis workflow. Transaction labels and category fields are used to summarize the data, and the resulting measures are presented through Power BI visuals. The available project information supports the following high-level stages:

- **Data review:** inspect the transaction fields used in the analysis, including class, amount category, time, and hour.
- **Preparation:** organize fields and categories so that they can be used consistently in summaries and visuals.
- **Metric calculation:** report transaction counts, fraud rate, and total fraud amount.
- **Visualization:** compare fraud and genuine activity across categories, hours, and alert levels.
- **Interactive exploration:** use the hour filter to examine dashboard results for selected transaction hours.

The exact data-cleaning operations, formulas, and any additional Python calculations should be confirmed against the included notebook before this report is treated as a complete technical specification.

6. Power BI Dashboard

The dashboard uses a dark fintech-inspired visual theme to create a clear monitoring interface. Its main components are:

Dashboard component	Purpose
Genuine vs Fraud Transactions	Compares the number of genuine and fraudulent transactions.
Fraud Rate by Transaction Amount Category	Compares fraud rates across Low, Medium, High, and Very high categories.
Fraud Transactions by Hour	Shows how fraudulent transaction counts vary by hour.
Transaction Distribution by Alert Level	Shows transaction counts across Normal, Low, Medium, and High alert levels.
Fraud Rate by Alert Level	Compares the fraud rate for each alert level.
Hour filter	Allows users to filter dashboard insights by transaction hour.

7. Key Results and Findings

7.1 Transaction composition

Of the 283,726 transactions reported, 473 are labeled as fraud and 283,253 as genuine. This means fraudulent transactions account for a small share of the total transaction volume.

7.2 Fraud rate by alert level

Alert level	Transaction count	Fraud rate
Normal	269,530	0.0282%
Low	13,564	0.4202%
Medium	363	31.4050%
High	269	84.0149%

The reported fraud rate rises across the alert levels, from 0.0282% for Normal to 84.0149% for High. This indicates that the alert-level grouping is strongly associated with the observed fraud labels in this dataset. It should not, by itself, be interpreted as proof that alert level causes fraud or as evidence of predictive performance.

7.3 Amount-category and hourly patterns

The dashboard includes a comparison of fraud rates across transaction amount categories and a chart of fraud counts by hour. These visuals support exploratory questions such as whether certain amount groups or hours show relatively more fraud activity. Specific category rankings and hourly peak values should be read directly from the dashboard, since the exact underlying chart values were not provided for this report.

8. Business Value

- Provides a compact view of fraud volume and overall fraud rate.
- Helps users compare fraud patterns across amount categories and hours.
- Highlights how observed fraud rates differ across alert levels.
- Makes key metrics easier to communicate to analysts and non-technical stakeholders.
- Supports exploratory monitoring and can help identify areas for deeper investigation.

9. Limitations

- The report summarizes the supplied dashboard metrics; the underlying dataset was not attached to this report.
- The Power BI .pbix file is not included in the repository due to its file size, so the dashboard cannot be independently refreshed from this repository alone.
- No predictive model, train/test split, confusion matrix, precision, recall, F1-score, or ROC-AUC results were supplied. Therefore, no machine-learning performance claim is made.
- The exact dataset source, observation period, currency/unit for the fraud amount, and detailed preprocessing steps should be documented from the notebook or dataset source.
- Observed relationships in the dashboard are descriptive and do not establish causation.

10. Recommendations and Future Scope

- Document the original dataset source, data dictionary, observation period, and any relevant currency or amount units.
- Add a data-quality section describing missing values, duplicates, data types, and preprocessing decisions.
- If the project is extended into predictive fraud detection, evaluate a suitable classification model using a time-aware or otherwise appropriate validation strategy.

- Because fraud is a rare event, report precision, recall, F1-score, PR-AUC, and confusion-matrix results rather than relying on accuracy alone.
- Consider cost-sensitive evaluation that reflects the different impacts of false positives and false negatives.
- Add a concise dashboard screenshot and a link to the notebook in the repository README.

11. Conclusion

The Credit Card Fraud Detection project demonstrates how transaction data can be summarized and communicated through Python-based analysis and an interactive Power BI dashboard. The reported dataset contains 283,726 transactions, of which 473 are labeled as fraudulent, producing an overall fraud rate of 0.1667%. The alert-level results show a substantial increase in observed fraud rate at higher alert levels. Together, the KPIs and visualizations provide a useful descriptive overview of transaction activity and fraud patterns.

The project can be strengthened further by documenting the complete data-preparation workflow, providing dataset provenance, and—if predictive detection is within scope—adding validated model evaluation.

12. Project Files

File	Description
README.md	Project overview, objectives, dashboard details, and repository guidance.
dashboard_preview.png	Static preview of the Power BI dashboard.
Credit_Card_Fraud_Analysis.ipynb	Jupyter Notebook containing the Python analysis.

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Project: Credit Card Fraud Detection Analysis
Tools: Python, Jupyter Notebook, Pandas, NumPy, Matplotlib, Seaborn, Microsoft Power BI, Power Query, and DAX